



grant adams - NOAA Federal <grant.adams@noaa.gov>

Maturity output

20 messages

Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Wed, Feb 25, 2026 at 11:10 AM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>

Hi Grant/Cody,

The array of p(molt) at size based on the new maturity workflow is attached. Let me know if you have any questions or would like anything different. I'd be happy to modify as you need!

Best,
Emily

--

Emily Ryznar*Research Fisheries Biologist**Kodiak Lab, Alaska Fisheries Science Center*

NOAA Fisheries | U.S. Department of Commerce

Office: (907) 308-3273

www.fisheries.noaa.gov

 SNOW_male_pmolt_array.csv
13K**cody szuwalski - NOAA Federal** <cody.szuwalski@noaa.gov>

Wed, Feb 25, 2026 at 11:11 AM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

hooray! Thanks much.

Cody

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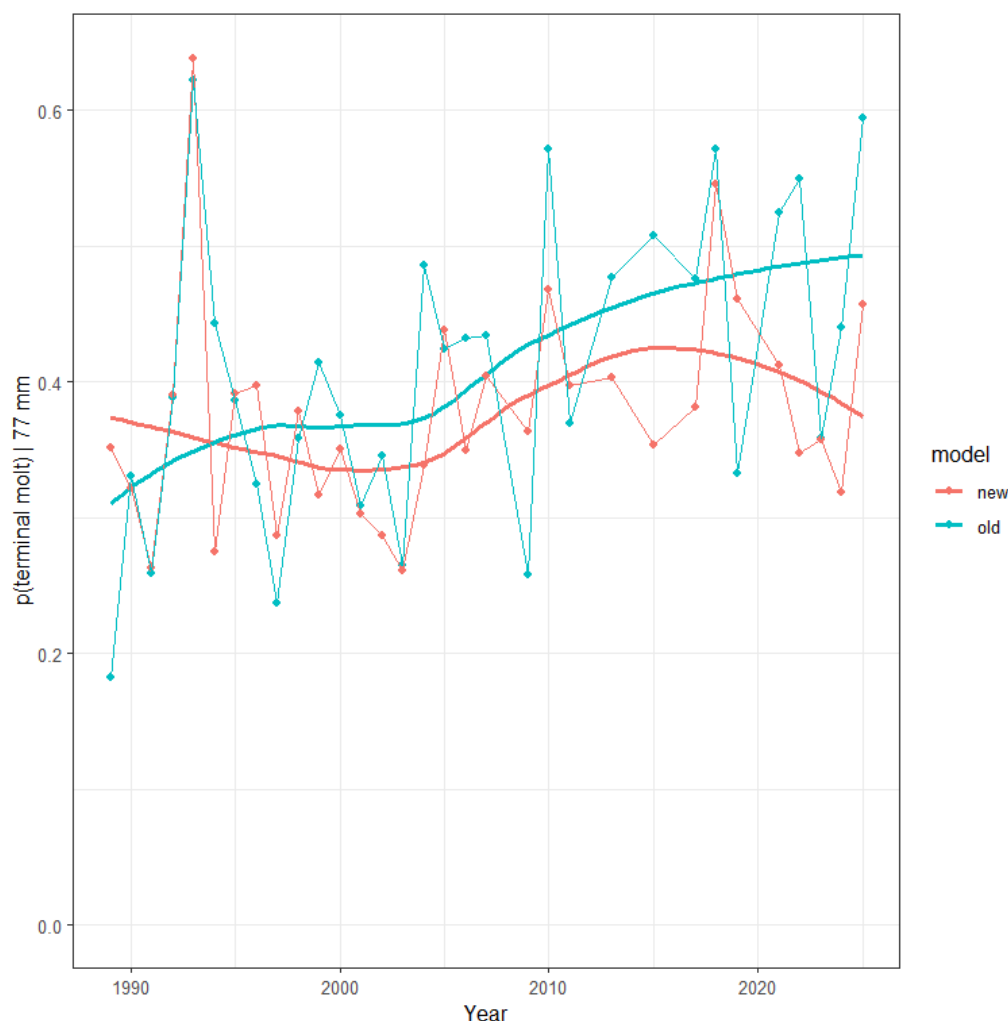
cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Wed, Feb 25, 2026 at 1:38 PM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

It's interesting how much more muted the trend in the probability of maturing (here at 77mm) is with your new methods.



Not really a trend at all anymore...

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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Wed, Feb 25, 2026 at 1:40 PM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Is your repo for this public?

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Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Wed, Feb 25, 2026 at 3:52 PM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Hi Cody,

Yep, the repo is public and can be found here: <https://github.com/eryznar/Chionoecetes.maturity.workflow>.

The array I sent was generated using the "make_stockassessment_inputs.R" script, which calls the function in "calc_maturepop_estimates_function.R". Let me know if anything isn't clear!

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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Wed, Feb 25, 2026 at 3:58 PM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thanks! Much appreciated. I'll report back soon.

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Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Wed, Feb 25, 2026 at 4:19 PM

Sounds good!

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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Wed, Feb 25, 2026 at 4:22 PM

Specifically, I want to see the map of the spatial effect for 75-80mm carapace width snow crab from 2022. Or even more specifically, the surface of probability that is multiplied by the spatial cpue data to get the final numbers of newly matured snow crab in that range from 2022. Are either of those saved anywhere?

75-80mm in 2022 is somewhat arbitrary, but it is one of the years and size bins that showed the largest difference between old and new. (not that I think old is 'right', just trying to track down differences and understand what is going on better!)

I'm curious at what the degrees of freedom consumed by estimating the spatial process are.

I'm trying to step through from beginning to end for a single size bin in a single year.

Thanks for whatever you can do in pointing me in the right direction!

Cody

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Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Wed, Feb 25, 2026 at 4:49 PM

I've attached a dummy script that should output what I think you're looking for, or at least give you a place to start. It also bypasses the model uncertainty estimation and just gives you point estimates of the probability. Hopefully this helps?

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 **script_for_cody.R**
2K

cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Wed, Feb 25, 2026 at 5:46 PM

thanks! I'll poke around and report back.

[Quoted text hidden]

cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thu, Feb 26, 2026 at 9:46 AM

Morning!

I'm getting this error in predict():

```
> preds <- predict(model, sub1, type = "response")

Error in getParameterOrder(data, parameters, new.env(), DLL = DLL) :
  Error when reading the variable: 'calc_weighted_avg'. Please check data and parameters.

In addition: Warning messages:
1: The installed version of sdmTMB is newer than the version that was used to fit this model. It is
possible new parameters have been added to the TMB model since you fit this model and that prediction
will fail. We recommend you fit and predict from an sdmTMB model with the same version.
2: In getParameterOrder(data, parameters, new.env(), DLL = DLL) :
  Expected scalar. Got length=0
3: In getParameterOrder(data, parameters, new.env(), DLL = DLL) :
  Expected object. Got NULL.
```

have you seen this before?

also do you know the version of sdmTMB you are using?

Thanks!

Cody

[Quoted text hidden]

cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
 To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
 Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thu, Feb 26, 2026 at 9:50 AM

Hm. nvm. new error after rerunning script. I will keep poking and report back.

Talk soon!

Cody

[Quoted text hidden]

Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
 To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
 Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thu, Feb 26, 2026 at 9:53 AM

Yep, I get that error when I try using the sdmTMB models on my VM because the models were fit on my PC (older R version) but my VM has a newer R version. It has been on my list to update R on my PC and refit the models, so this is a good push to do so! It shouldn't take very long, and I can let you know when I'm done.

Sorry about that!

[Quoted text hidden]

cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
 To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
 Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thu, Feb 26, 2026 at 9:56 AM

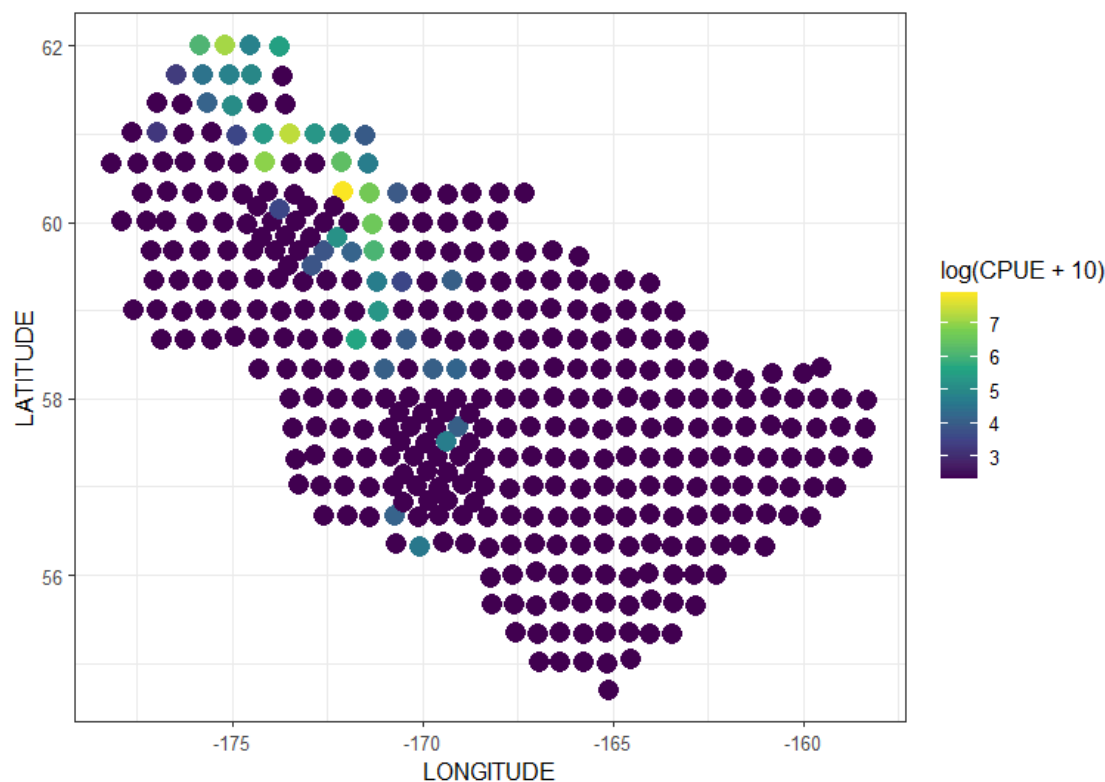
sweet! no big hurry. does the plot of the surface at the end of the script you sent output for you? if so, would you mind copy pasting here? if not, no biggie either! I can wait.

[Quoted text hidden]

Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>
 To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>
 Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Thu, Feb 26, 2026 at 10:26 AM

It does! But it's just the predicted mature CPUE 75-80 at each station in 2022, not a continuous surface.



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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Thu, Feb 26, 2026 at 10:29 AM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

got it. I'm interested in the surface of probability mature that is applied to the cpue data to split them mature/immature. is that easy to pull? (sorry for so many requests! if it's tricky, no worries)

[Quoted text hidden]

Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Thu, Feb 26, 2026 at 10:54 AM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>

Ahhh gotcha, I think I know what you mean! I'll get back to you.

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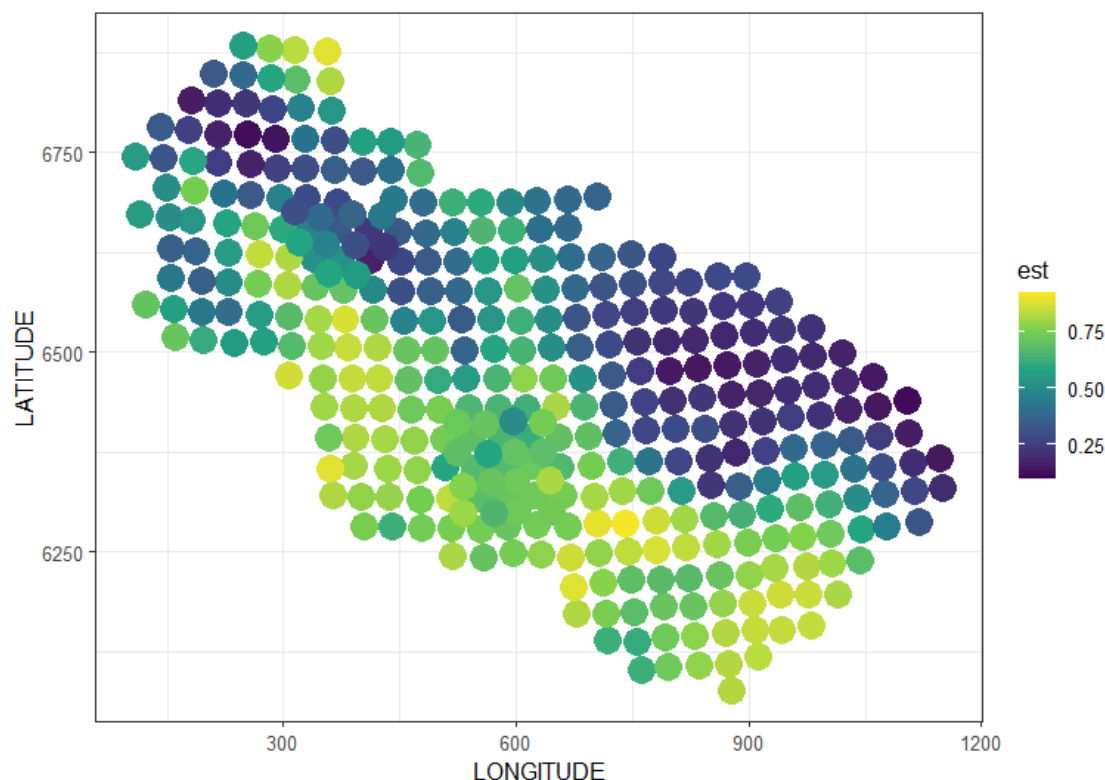
Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Thu, Feb 26, 2026 at 11:19 AM

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Something like this?



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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Thu, Feb 26, 2026 at 12:06 PM

To: Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

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yes! thank you. I'm trying to wrap my head around how to justify what amount of variability in spatial difference in estimated probability is warranted given the data. for this size range and year, most survey stations don't have chela measurements, those that do mostly have 1 measured chela, with a maximum of four (I think four?...need to look again). I will share some figures and thoughts soon. how did you think about this question when designing your analyses?

I appreciate your help!

Cody

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Emily Ryznar - NOAA Federal <emily.ryznar@noaa.gov>

Thu, Feb 26, 2026 at 1:10 PM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

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I think the point you make is exactly where I envisioned a spatiotemporal approach would be most beneficial by borrowing strength across space and time for size bins with poor chela sample sizes in some years and propagating uncertainty in model estimation due to those low sample sizes (and spatiotemporal variability). For example, that size range has the widest uncertainty for 2022 in the ogives. I don't ever use the entire predicted spatial surface of the probability because of extrapolation beyond our actual measured crab in each year of the survey and only use crab that we actually measured. I definitely recognize there are some severe limitations to our chela data, and we determined this form of analysis would address those limitations in the most robust way.

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